Biomass CHP — Maker Shortlist

Potential manufacturers for the Mitsue biomass CHP, split by the two build stages: **Tier A — a 10–20 kWe first prototype**, and **Tier B — the 2 × 0.6 MWe (1.2 MWe) commercial fleet**.

Policy reminder: the *commercial* units must be a **Japanese manufacturer** (domestic service/parts, simpler FIT + 交付金 paperwork, local supply-chain story). Imported units are **benchmark-only, not for procurement** — with one possible, explicitly-flagged exception at prototype scale (see Tier A).

Update 2026-10-15 (Rob: Japanese CHP options only):

- **ENTRENCO is dropped from consideration.** WegscheidEntrenco GmbH (Regensburg, HRB 17110) entered insolvency on 2026-04-30 (Amtsgericht Regensburg, 4 IN 133/26; administrator RA Michael Verken; creditors' meeting 2026-07-23 to decide shutdown vs continuation, outcome not checked; source: insolvenz-radar.de). Its Japan units (5 installed) face supplier, parts and warranty risk. Its predecessor-technology firm Entrade Energiesysteme AG was insolvent in 2018.
- Foreign makers (ESPE, Esperia, All Power Labs, Spanner) stay **benchmark only**. **Volter (Finland) is kept as an active contact** at Rob's request (2026-10-15): the Finland connection matters to him; kiri gasification inquiry drafted to sales@volter.fi.
- **Conflict to resolve:** the JWBA 2024 list (see memory `reference_jwba_equipment_database`) shows a Neonite **HKW50, up to 55 kWe, ≤15% moisture with dryer, installed in Kumamoto**, which contradicts "Neonite smallest ~350 kWe" below. If the HKW50 is real, a domestic gasifier exists at ~50 kWe, which would change the "no domestic gasifier at prototype scale" finding. Verify with Neonite.
- Kiri gasification inquiry: send to the Japanese makers (Neonite, Shizuoka Seiki) and the Hokkaido FPRI; see outreach draft. **Update 2026-10-16: Shizuoka Seiki declined — small gasification development discontinued.**

Authoritative reference: the JWBA list *国内で販売されている小規模木質バイオマス発電機器の一覧* catalogues every domestic unit with specs — use it to confirm any figure below.

> ⚠️ All kW ratings here are **indicative / to verify** against the JWBA list or a vendor quote before use. CHP sizing in the wider project is still a planning estimate.

The key finding — the prototype scale has no domestic gasifier

Domestic woody-**gasification** CHP has a floor around **~200–350 kWe** (Neonite's smallest is ~350 kWe; Shizuoka Seiki ~200 kWe). A 10–20 kWe prototype is an **order of magnitude below that floor**, so there is **no domestic-maker, gasification unit at prototype scale.**

This forces an explicit choice for the prototype (a decision, not a spec gap):

- **(a) Domestic non-gasification micro-CHP** — a wood-chip **boiler + Stirling engine** at ~10 kWe *does* exist domestically (巴商会). Keeps the Japanese-maker policy intact; trades gasification for Stirling/combustion at this stage.
- **(b) Imported micro-gasifier as a one-off pilot exception** — All Power Labs / Volter / Spanner HKA-class. Matches the eventual gasification tech, but breaks the domestic-maker policy for the prototype only — would need to be justified as pilot-only.
- **(c) Research / university partnership build** — slower, but keeps IP and learning local.

Recommendation to decide: **(a) for a fast, policy-clean, revenue-visible prototype**, or **(b) if the prototype's main job is to de-risk the exact gasification tech** we scale to.

Tier A — 10–20 kWe first prototype

Maker	Tech / scale	Domestic?	Price	Notes
巴商会 (Tomoe Shokai)	wood-chip boiler + **Stirling engine**, ~10 kWe / ~180 kWth	✅ (distributor)	Not published — quote-only	Closest domestic match to prototype scale; combustion+Stirling, *not* gasification. Lead for option (a).
ネオナイト (Neonite)	downdraft 2-stage **gasification**, smallest ~350 kWe	✅	Not published	Too big for a prototype — belongs to Tier B. Biochar/wood-tar recovery.
静岡製機 (Shizuoka Seiki)	small outdoor gasification, ~200 kWe	✅	Not published	**DISCONTINUED (2026-10-16): Shizuoka Seiki replied twice (inquiry desk + international desk, 石黒) that it has stopped developing small gasification power systems, withdrawn from the business and disposed of the equipment.** Remove from candidates.
All Power Labs (Power Pallet)	downdraft micro-**gasification**, ~10–25 kWe	❌ US (on JWBA list)	Historically listed ~$40,000–50,000 on their own site — **not re-verified this session, confirm on apl.solar before citing**	Best *gasification* size match; **import → pilot-exception only**.
Volter	CHP micro-gasifier, ~40 kWe	❌ Finland	Not published	Benchmark; import-only.
Spanner Re² (HKA-9 / HKA-30 / HKA-	wood gasifier-CHP modules, ~9–45	❌ Germany	Not published	Benchmark; smallest module

Maker	Tech / scale	Domestic?	Price	Notes
45)	kWe			near prototype scale; import-only.

⚠️ **Price caveat:** none of the domestic makers (巴商会/ネオナイト/静岡製機/中外炉/神鋼) publish list prices — this is a quote-only industry (site conditions, fuel spec, grid connection, and heat-offtake design all move the number). The only figures found with any public backing are the imported micro- gasifiers above, and even those are indicative, not confirmed current quotes. **Do not present the prices in this doc as firm — they are a planning placeholder until vendor RFQs come back**, per the outreach already drafted in `mitsue_biomass_visit_request_emails.md`.

Tier B — 2 × 0.6 MWe (1.2 MWe) commercial fleet

Small woody-gasification tops out ~200–500 kWe, so 0.6 MWe per line is a **cluster of modules**. (Reproduced from `mitsue_forest_workforce_energy_plan.md` §5.)

Maker	Tech / scale	Fit	Price	Notes
中外炉工業 (Chugai Ro)	rotary-kiln gasification cogen	✅ **lead**	Not published	Since 1997, NEDO-backed; reaches our per-unit size; tolerates rough fuel (branches/bark).
神鋼環境ソリューション (Shinko / Kobe Steel grp)	~500 kWe-class up	backup	Not published	⚠️ confirm **gasification** vs steam (big plants are combustion).
~~静岡製機 (Shizuoka Seiki)~~ **(development stopped, 2026-10-16)**	small ~200 kWe outdoor unit	cluster	Not published	METI-funded, aimed at agriculture/forestry — closest in spirit.
ネオナイト (Neonite)	downdraft two-stage, ~350 / 750 kWe	cluster	Not published	Recovers charcoal + wood tar (biochar angle).

The only hard cost figure found in public sources is operating (not capital) cost: 中外炉's own published paper on small-scale gasification puts **generation cost at ~¥5/kWh**, meaning even a ¥40/kWh FIT tariff nets a thin margin — no installed-cost (¥/kW capex) figure is published anywhere for these units. This reinforces [[reference_fit_grid_biomass]]'s "compute >> FIP" finding: margin depends on a high-value offtake (the DC load), not the FIT rate. **No capex number should be quoted to Mizuho/investors until a vendor RFQ comes back** — treat any capex figure elsewhere in this project as a placeholder.

> **Imported — benchmark-only, not for procurement:** FORTES Energy (Latvia; 0.14–1.2 MWe gasification CHP, new Tokyo entity), Spanner Re² and Burkhardt (Germany; e.g. V3.90 module). Useful for spec comparison; do not meet the Japanese-manufacturer requirement.

Prototype siting (updated 2026-09-27 — supersedes 2026-07-17 decision)

Site the prototype **directly at the Local Forestry Partner worksite on the hill (牛峠工場, 神末797)** — both the chip drying step and the CHP unit itself, not the earlier plan of siting the CHP at Sugano Organic with chips trucked down from the hill. Rationale: eliminates the ~2km chip haul entirely (fuel never leaves the forestry site), and lets the CHP's own waste heat feed the drying step on-premise (matches the disaster-CHP paper's finding — see `eis_53-1_2024_wood_biomass_chp_disaster_EN_translation.md` §2.2.3/Eq.② — that on-site drying draws directly on CHP thermal output, so co-locating drying + generation avoids a second haul AND keeps that heat loop short). This also converges with the "later commercial CHP sits at 神末" plan from the three-node siting note — the prototype now previews the commercial site's configuration instead of testing a different (Sugano) location.

Consequence for fuel/moisture, per the CHP maker table above: since fuel arrives fresh/green straight from harvest (not pre-dried at a mill), the choice is between (a) a standalone dryer loop ahead of a low-moisture-spec unit (Volter ≤15%, Entrenco ≤12%), or (b) a wet-tolerant packaged unit (ESPE ≤45%, Esperia ≤40%) that folds drying into the CHP package itself — this now needs to be decided as part of unit selection, not assumed away. Fuel-supply-rate question (harvest tonnage available at Local Forestry Partner) remains open per [[project_sugano_fuel_partner]]'s 2026-09-03 update — Sugano/Niwa fuel-chipper route is a separate question, not this pilot's fuel source.

(A public prototype demo can still run separately at the Koryukan / an unused factory if wanted for visibility, but the working pilot itself sits at Local Forestry Partner.)

Fuel chippers — sized to each CHP tier

Fuel spec: **non-construction-grade wood only** — thinnings, branches, bark, slabwood, reject logs (see [[project_sugano_fuel_partner]]'s cascading-use model with 丹羽製材, whose *existing* chipper makes pulp/paper chips and is NG for fuel — a separate fuel-spec chipper is needed either way). Fuel-grade chip spec: <20% moisture after drying, no strict size-grading (unlike pulp chips).

Tonnage benchmark: [[reference_mishima_fukushima_chp]]'s scenario figure (simulation, not an installed unit — NIES confirmed 2026-09-07) is **~750 t/yr per 50 kWe**.

Real operating anchor found: Spa Hotel Abukuma (西郷村, Fukushima) — Entrade Energiesysteme (Germany) E3 (25 kWe) + WE-50P (49.9 kWe) units, running since Nov 2018, measured ratio ~1 kWh elec + 2 kWth heat per 1 kg pellet. Imported (benchmark-only per policy above), but real measured output data, unlike Mishima's simulation. Capex/fuel-tonnage not yet public — inquiry drafted, see `mitsue_email_fujitakk_abukuma_chp_inquiry.md`.

CHP tier	Approx. fuel demand	Chipper class needed	Example models found	Price
Tier A prototype (10–20 kWe)	~150–300 t/yr	Small tractor-mount / PTO or hand-fed	富士テックス GS75G–GS126G (小型), or Kyoritz/Yamamoto small self-propelled	Small PTO unit (e.g. MKW-100, 18 HP tractor-mount): **¥390,000 new**. Small self-propelled diesel (used market): **¥400,000–**

CHP tier	Approx. fuel demand	Chipper class needed	Example models found	Price
				650,000 (Yahoo Auctions closed-lot range). New small self-propelled: no confirmed list price found — expect roughly ¥1–3M range based on used-market floor, **unconfirmed, get a dealer quote**.
Tier B commercial (1.2 MWe, ~18,000 t/yr scaled from Mishima ratio)	~15,000–18,000 t/yr	Mid-size self-propelled (continuous-duty, ~100–150 HP class)	富士テックス GS133GH / GS152GH / GS285D (中型) — no public price found	Not published — **request quote from Fujitex/フジテックス directly**; do not guess a figure for the RFQ stage.
Reference — industrial continuous-feed class (for scale comparison, not a Mitsue-size recommendation)	200–380 m³/h capacity	Track-mounted drum chippers	Morbark 4036 Track (200 m³/h, 765–800 HP, 25.8 t), Komptech Axtor (300 m³/h, 23 t), Pezzolato drum (up to 380 m³/h)	No prices published (imported industrial capital equipment, quote-only); shown only to illustrate that Tier B's tonnage does **not** require this class — a mid-size Japanese unit run on a normal work schedule (not 24/7) covers it.

Bottom line on chipper price data: like the CHP units themselves, essentially no manufacturer publishes a chipper list price in Japan — confirmed for Fujitex, Kyoritz/Yamabiko, and all imported industrial brands searched. The only two real figures found are the ¥390,000 tractor-mount unit and the ¥400–650k used-market range for small self-propelled units. **Next action: request quotes from Fujitex (中型 GS133GH/GS152GH) and confirm with 丹羽製材/Niwa-san whether a shared fuel chipper at Tier-B scale is realistic for their yard, or whether it should sit at 牛峠工場 instead.**

Next steps

- Pull the current **JWBA small-scale list** and confirm every kW figure above.
- Decide the prototype path — option (a) / (b) / (c) — before requesting quotes.
- Confirm **Sugano Organic site readiness** (space, siting consent from Tokuo Aomi) and the chip-supply handshake with 丹羽製材 / the 牛峠工場 chipper.

- Vendor/site-visit outreach already drafted in `mitsue_biomass_visit_request_emails.md` (中外炉 / 神鋼 / 静岡製機 / ネオナイト priority; Spanner / Burkhardt as imported benchmarks).

Sources

- JWBA 小規模木質バイオマス発電機器の一覧 — https://jwba.or.jp/database/list-small-woody-biomass-generation/
- JWBA list PDF (alphabetical, incl. All Power Labs) — https://jwba.or.jp/wp/wp-content/uploads/2024/11/20241122_国内で販売されている小規模木質バイオマス発電機器の一覧.pdf
- 巴商会 木質バイオマスボイラ＋発電（Stirling, ~10 kWe/180 kWth） — https://tomoeshokai.com/product/woody_biomass_boiler/generator
- ネオナイト 小規模木質バイオマスガス化発電（350/750 kW） — http://www.neonite.jp/biomass_lp/
- Metoree バイオマス発電機メーカー一覧 — https://metoree.com/categories/2730/
- 中外炉 小規模ガス化発電コスト論文（笹内, ~¥5/kWh 発電コスト） — https://www.npobin.net/research/data/155thSasauchi.pdf
- JWBA 木質破砕機・チッパー一覧 — https://jwba.or.jp/database/list-chipper/
- JWBA チッパー仕様一覧 PDF — https://jwba.or.jp/wp/wp-content/uploads/2022/04/木材破砕機_チッパーの一覧-破砕チップ対応_20190510.pdf
- フジテックス 木材チッパー製品一覧 — https://www.fjtex.co.jp/kankyo/products/tipper/
- Forest Journal「木材破砕機・2023年チッパー5選」（Morbark/Doppstadt/ERJO/Komptech/Pezzolato spec) — https://forest-journal.jp/tools/38595/
- MKW-100 トラクター装着チッパー ¥390,000 — https://chuko-noki.com/704/

Related docs

- `mitsue_forest_workforce_energy_plan.md` §5 (energy stack) — CHP maker table source
- `mitsue_biomass_visit_request_emails.md` — vendor visit outreach